TP GUIDED: Implementation of a CI/CD Pipeline with GitHub Actions

Students : 2SCL / BI

1. OBJECTIVE

The objective of this lab is to implement a simple CI/CD pipeline for a static web application using GitHub Actions. The student will learn continuous integration, continuous testing, artifact generation, and deployment using GitHub Pages.

2. PREREQUISITES

- A GitHub account
- Git installed on local machine
- Basic knowledge of HTML, CSS, JavaScript
- Basic Git commands (add, commit, push)
- A code editor such as VS Code

3. PROJECT SETUP

Step 1: Create project folder

Step 2: Create application files

index.html:
- Simple HTML page with a button and script reference

C: > Users > hp > Desktop > CI-CD BI-2SCL > index.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>My First Web App</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <h1>Simple CI/CD Demo</h1>

    <button onclick="showMessage()">
        Click Me
    </button>

    <p id="message"></p>

    <script src="app.js"></script>

</body>
</html>
```

app.js:

Casa.cfg | index.html | style.css | JS app.js

C: > Users > hp > Desktop > CI-CD BI-2SCL > JS app.js > showMessage

```javascript
function showMessage() {
    document.getElementById("message").innerHTML =
        "CI/CD with GitHub Actions works version 2!";
}
```

style.css (optional):
- Basic styling for the page

C: > Users > hp > Desktop > CI-CD BI-2SCL > # style.css > button

```css
body {
    font-family: Arial, sans-serif;
    text-align: center;
    margin-top: 100px;
}

button {
    padding: 10px 20px;
    font-size: 16px;
}
```

4. INITIALIZE GIT AND CONNECT TO GITHUB

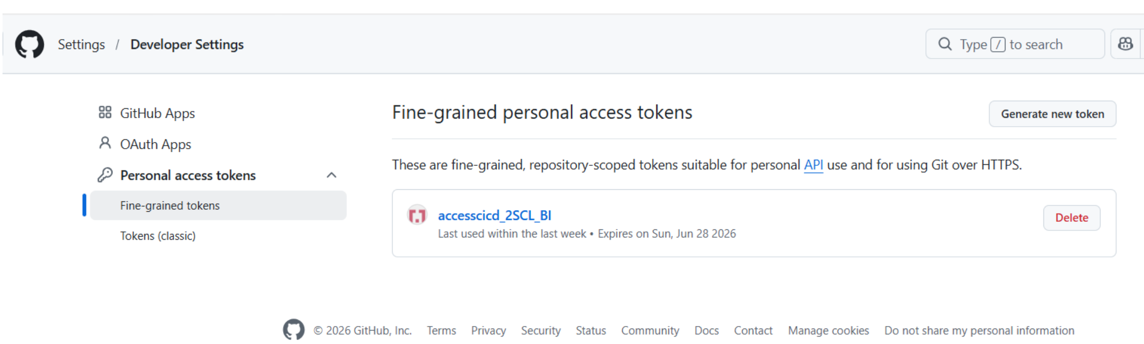
```
git init
git add .
git commit -m "Initial commit"
git remote add origin <your-repo-url>
git branch -M main
git push -u origin main
```

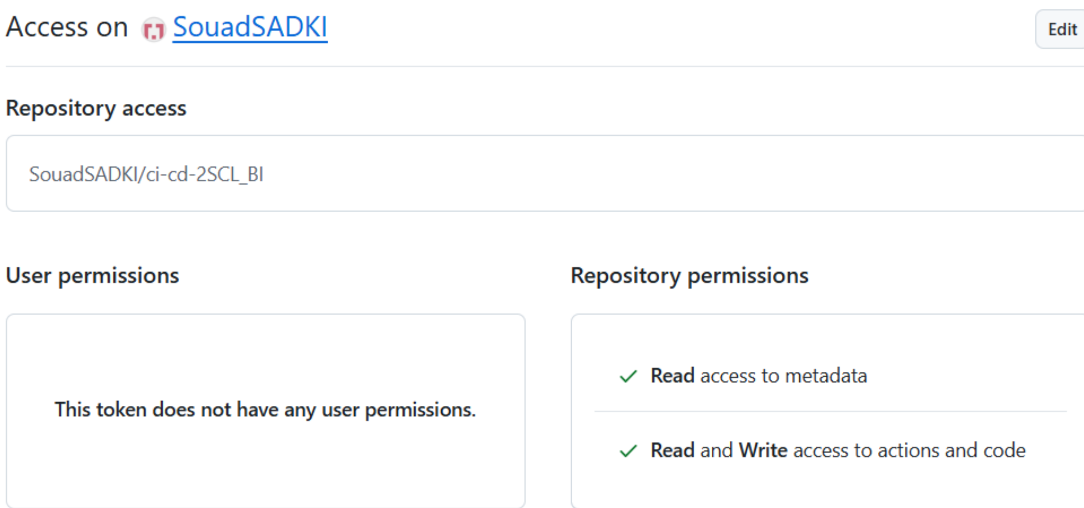
```
hp@DESKTOP-A0NOD3R MINGW64 ~
$ cd Desktop/

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop
$ cd CI
CI-CD BI-2SCL/            CISCO_LABS/            Cisco Packet Tracer.lnk

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop
$ cd CI-CD\ BI-2SCL/

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL
$ git init
Initialized empty Git repository in C:/Users/hp/Desktop/CI-CD BI-2SCL/.git/

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git add .

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git commit -m "change1"
[master (root-commit) 43ab04c] change1
 3 files changed, 34 insertions(+)
 create mode 100644 app.js
 create mode 100644 index.html
 create mode 100644 style.css

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git remote add origin https://github.com/SouadSADKI/ci-cd-2SCL_BI.git
```

Then Push:

```
hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git push -u origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 711 bytes | 237.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SouadSADKI/ci-cd-2SCL_BI.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

TIP : ACCESS TOKEN FOR THE REMOTE REPO





5. CI/CD PIPELINE IMPLEMENTATION (GITHUB ACTIONS)

Create the following file:
.github/workflows/ci-cd.yml

ci-cd-2SCL_BI / .github / workflows / ci-cd.yml

SouadSADKI Update ci-cd.yml ✕

Code Blame 98 lines (78 loc) · 1.95 KB

```yaml
name: Simple CI/CD Pipeline

on:
  push:
    branches:
      - master

jobs:

  # ====================
  # 1. BUILD STAGE
  # ====================
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Validate project structure
        run: |
          echo "Checking files..."
          test -f index.html
          test -f style.css
          test -f app.js
          echo "Build stage completed successfully!"
```

```yaml
  # =====================
  # 2. TEST STAGE
  # =====================
  test:
    runs-on: ubuntu-latest
    needs: build   # runs after build

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Basic tests
        run: |
          echo "Running simple tests..."

          grep -q "<html>" index.html
          grep -q "function" app.js

          echo "All tests passed!"

  # =====================
  # 3. PACKAGE STAGE
  # =====================
  package:
    runs-on: ubuntu-latest
    needs: test

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Create deployment package
```

```yaml
        run: |
          mkdir build
          cp index.html build/
          cp style.css build/
          cp app.js build/
          echo "Package created"

      - name: Upload artifact
        uses: actions/upload-artifact@v4
        with:
          name: web-app
          path: build/

  # =====================
  # 4. DEPLOY STAGE (GitHub Pages)
  # =====================
  deploy:
    runs-on: ubuntu-latest
    needs: package

    permissions:
      contents: read
      pages: write
      id-token: write

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Setup GitHub Pages
        uses: actions/configure-pages@v5

      - name: Upload Pages artifact
        uses: actions/upload-pages-artifact@v3
        with:
          path: .

      - name: Deploy to GitHub Pages
        uses: actions/deploy-pages@v4
```

Pipeline stages:

BUILD STAGE
- Checkout code
- Verify required files exist (index.html, app.js)

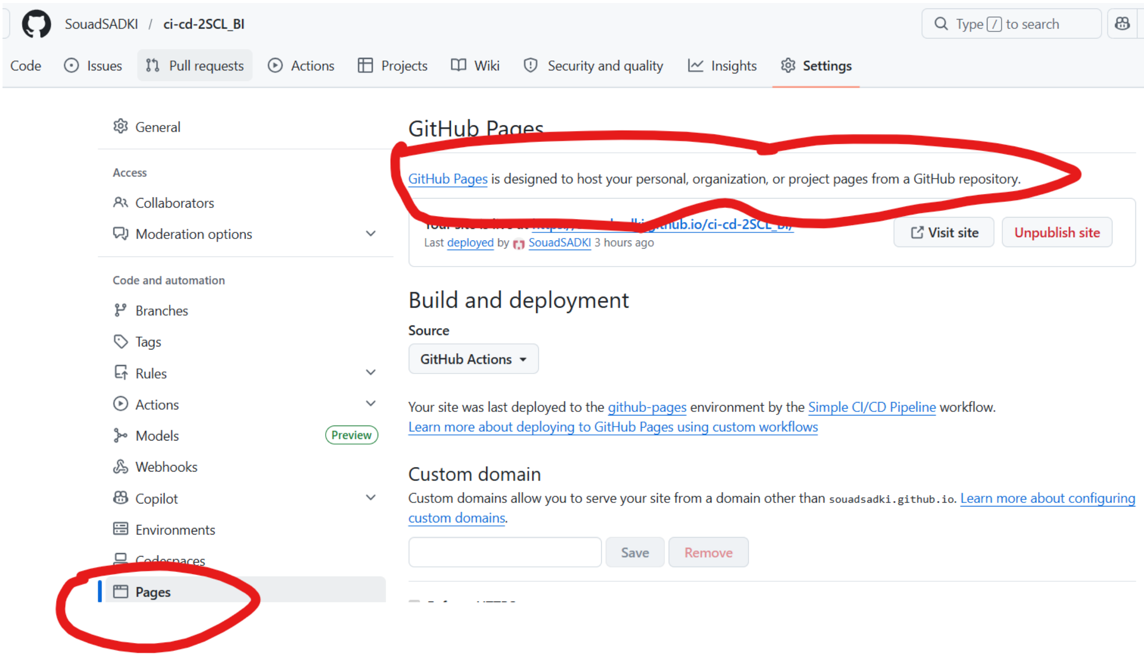
TEST STAGE
- Validate basic structure of HTML and JS files using simple grep checks

PACKAGE STAGE
- Create a build folder
- Copy application files into build directory
- Upload build artifact

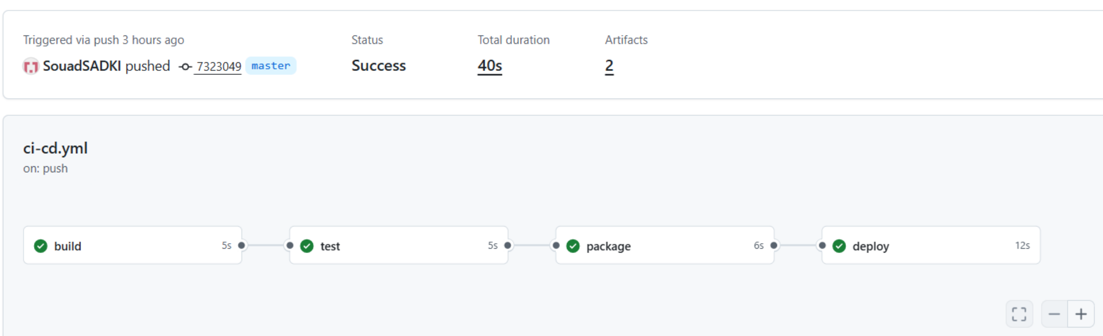
6. WORKFLOW EXECUTION
After pushing the workflow file:
- GitHub Actions automatically triggers on push to main branch
- Each job runs sequentially based on dependencies
- Build must pass before test runs
- Test must pass before packaging

7. DEPLOYMENT WITH GITHUB PAGES

Steps:
- Go to repository settings
- Navigate to Pages section
- Select GitHub Actions as deployment source
- Add deploy job in workflow using actions/configure-pages and deploy-pages



8. FINAL RESULT
After successful execution:
- CI pipeline validates code automatically
- CD pipeline deploys application
- Application becomes accessible via GitHub Pages URL



Test our pipepline :

1. Modify the HTML file

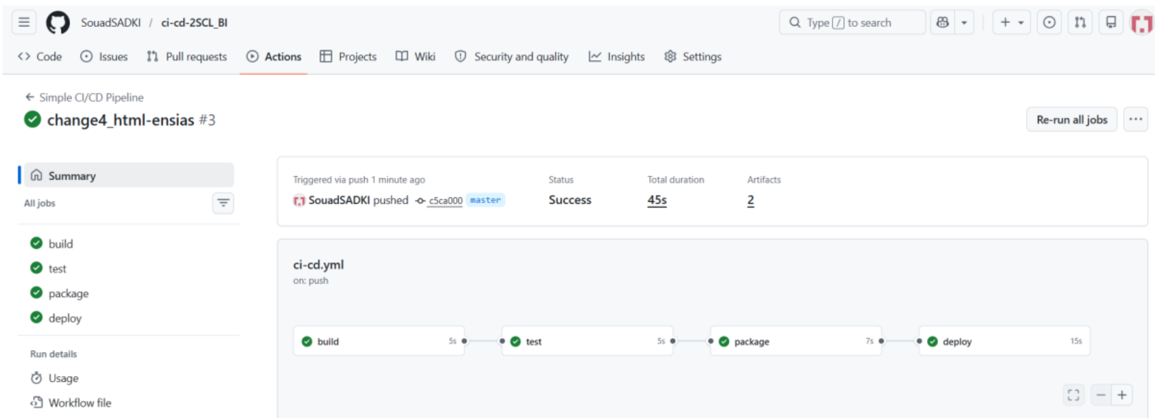
C: > Users > hp > Desktop > CI-CD BI-2SCL > index.html > html > head > title

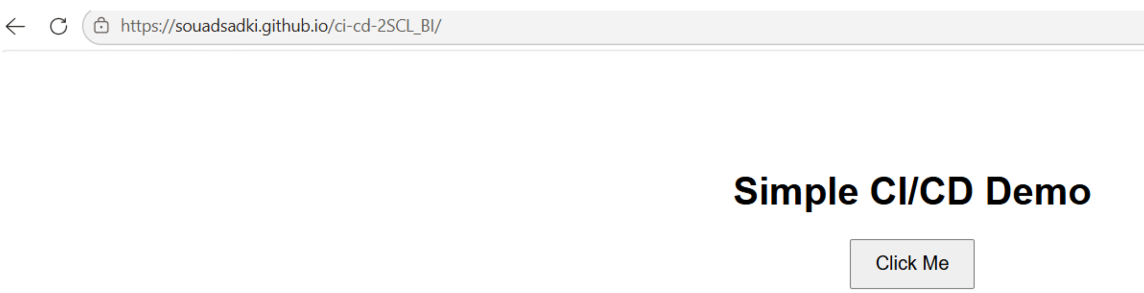
```html
<!DOCTYPE html>
<html>
<head>
    <title>My First Web App at ENSIAS </title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

    <h1>Simple CI/CD Demo</h1>

    <button onclick="showMessage()">
        Click Me
    </button>

    <p id="message"></p>

    <script src="app.js"></script>

</body>
</html>
```

Push the new verison oof the HTML file :

```
hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git add .

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git commit -m "change4_html-ensias"
[master c5ca000] change4_html-ensias
 1 file changed, 1 insertion(+), 1 deletion(-)

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 303 bytes | 303.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/SouadSADKI/ci-cd-2SCL_BI.git
   7323049..c5ca000  master -> master

hp@DESKTOP-A0NOD3R MINGW64 ~/Desktop/CI-CD BI-2SCL (master)
$ 
```

IS our pipeline going to detect this change :





Let's change the background of our html page :

C: > Users > hp > Desktop > CI-CD BI-2SCL > index.html > html > head

```html
<!DOCTYPE html>
<html>
<head>
    <title>My First Web App at ENSIAS </title>
    <link rel="stylesheet" href="style.css">
</head>
<body style="background-color: green;">

    <h1>Simple CI/CD Demo</h1>

    <button onclick="showMessage()">
        Click Me
    </button>

    <p id="message"></p>

    <script src="app.js"></script>

</body>
</html>
```

The results is :

CONCLUSION

This lab demonstrates a complete CI/CD workflow using GitHub Actions for a simple web application. It introduces automation principles used in modern DevOps pipelines.